

Lorentz invariance

Let X be a space-like vector. There exists a Lorentz transformation Λ such that the time component vanishes in the transformed vector $X' = \Lambda X$.

For example, let X be the space-like vector

$$X = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix}$$

Verify X is space-like.

$$X^\mu X_\mu = -3$$

We seek Λ such that

$$X' = \Lambda X = \begin{pmatrix} \cosh \eta & 0 & 0 & \sinh \eta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \eta & 0 & 0 & \cosh \eta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ z' \end{pmatrix}$$

This reduces to finding η such that

$$\cosh \eta + 2 \sinh \eta = 0$$

Solve numerically to obtain

$$\eta = -0.549306$$

Hence

$$\Lambda = \begin{pmatrix} 1.1547 & 0 & 0 & -0.57735 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -0.57735 & 0 & 0 & 1.1547 \end{pmatrix}$$

Eigenmath script